

**AMENDMENT NO. 1 OCTOBER 2004
TO
IS 715 : 2002 COATED ABRASIVES —
SPECIFICATION
(Fourth Revision)**

(Page 5, clause 9) — Substitute the following for the existing:

9 PRODUCT EVOLUTION AND TESTING

Product shall be tested according to type of bond described in 9.1, 9.2, 9.3, 9.4, 9.5, 9.6 and 9.7.

(Page 5, clause 9.6) — Insert the following clause after 9.6:

9.7 Determination of Aluminium Oxide

The procedure for determining aluminium oxide and combined iron oxide shall be as given in Annex H.

(Page 9, Annex G) — Insert the following Annex at the end:

ANNEX H

(Clause 9.7)

DETERMINATION OF ALUMINIUM OXIDE IN ABRASIVES

H-1 METHOD

H-1.1 Crush the sample to a finely divided powder in an agate pestle and mortar and weigh out approximately 0.3 g of the sample in a platinum crucible. Mix thoroughly with about ten times its weight of crushed potassium bisulphate, taking care not to lose any material during mixing.

H-1.2 Fuse the mixture in an ordinary Bunsen burner in a completely oxidizing flame, first slowly, till the frothing ceases and then vigorously for about half an hour, till a clear melt is obtained. Allow the mass to cool and take up the melt in 100 ml of hot distilled water containing 20 ml of hydrochloric acid in a beaker. Heat to boil and allow it to settle till the silica settles at the bottom. Filter through paper pulp in a 250-ml graduated flask. Wash the residue thoroughly several times with hot water, collecting the filtrate.

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H-1.3 Make up the filtrate to a volume of 250-ml by adding water. Take an aliquot portion, 50 ml in a 250-ml beaker and add water to make the volume approximately to 100 ml. Add about five grams of ammonium chloride and then ammonia until the solution is slightly alkaline. Boil continuously for about five minutes and allow the precipitate to settle. Filter through No. 31 Whatman filter paper and wash the residue thoroughly with the hot water containing a little of ammonium nitrate. Ignite the residue to constant weight in a weighed platinum crucible. Reweigh the contents. The difference gives the combined weight of iron oxide (Fe_2O_3) + aluminium oxide (Al_2O_3) - total oxide (R_2O_3). Take another 100 ml of the solution in a 250-ml beaker and heat to boil. As the solution begins to boil add drop by drop stannous chloride solution till the yellow colour of the solution changes to colourless one. Allow it to cool in ice and to the ice cooled solution add 10 ml of mercuric chloride solution and stir it well and keep it for five minutes A silky precipitate is formed. Make the solution to about 200 ml with cold distilled water and titrate with N/20 Standard potassium permanganate solution to a permanent pink colour, adding a few millilitres of phosphoric acid. From the amount of standard solutions used, calculate the percentage of iron oxide as given in the following formula:

$$\text{Iron oxide (Fe}_2\text{O}_3) = \frac{\text{Millilitres of potassium permanganate used} \times \text{Strength of the potassium permanganate} \times 250 \times 0.07984}{\text{Weight of the sample}}$$

The difference between total oxide and iron oxide gives the weight of the aluminium oxide.

NOTE — The above method is for the determination of Al_2O_3 which may exist as free alumina or in a combination with other molecules.

(MGPD 09)